

Architectural Justification - Podcast Guest Scheduling & Outline Builder

Lab 3 - Component Modelling & Architectural Pattern Selection

ARCHITECTURE SELECTED: LAYERED ARCHITECTURE

Architectural Choice

The system is organized into three layers: Presentation, Business, and Data. The Presentation layer contains the Booking UI and Host Dashboard. The Business layer contains the Scheduling Manager and Production Sheet Generator. The Data layer contains the Database Component. This structure matches the request-process-store flow of the scheduling system while keeping the design simple and modular.

Reason 1 - Clear Separation of Concerns

Guest and host interfaces are separated from core scheduling logic. Booking validation, slot-conflict checks, availability management, and rescheduling are handled by the Scheduling Manager, while production-sheet compilation and PDF generation are handled by the Production Sheet Generator. Changes to the user interfaces therefore do not require changes to the underlying business rules.

Reason 2 - Centralized Scheduling Consistency

Guest booking and host scheduling operations pass through the Scheduling Manager. This centralizes availability checks, booking validation, and rescheduling rules so that conflicting or unavailable slots are handled consistently regardless of which interface initiates the operation. It also supports the Lab 1 requirement that a slot must not be booked when it is unavailable.

Security Advantage

The Presentation layer does not access the Database Component directly. Requests are handled through the Business layer, where authorization and validation rules can be enforced centrally - for example, guests can be restricted to their own booking information while hosts can be granted availability-management and booking-management operations. This reduces the risk of inconsistent access-control checks across user interfaces.

Performance Benefit

For a system of this scale, a layered design avoids the additional network communication and serialization overhead that a microservices deployment would introduce between booking, scheduling, production-sheet, and storage functions. Business components can communicate with the database through local service interfaces, keeping common operations such as availability checks and topic retrieval low-latency.

Component / Interface Summary

The diagram contains five components and five interface connections: BookingService (Booking UI -> Scheduling Manager), HostManagementService (Host Dashboard -> Scheduling Manager), ProductionSheetService (Scheduling Manager -> Production Sheet Generator), and SQL-based persistence/topic-data interfaces from the business components to the Database Component. A dashed <<use>> dependency identifies the Scheduling Manager as the client of the Production Sheet Generator.

Design note: The corrected diagram avoids a direct Booking UI -> Production Sheet Generator dependency. Production-sheet generation and PDF download are handled through the business layer, preserving the layered separation required by the architecture.